

Detailed Feature Catalogue for the Current GBT+JEPA Models

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Abstract

This document catalogues the feature blocks used by the current system. It is intentionally more detailed than a summary report: it explains the economic meaning of the key engineered features, the state/input split used by XInputJEPA, the 42 JEPA features appended downstream, and the option-candidate features used by the 0DTE research models. The machine-readable source of truth remains the model artifacts and feature-name JSON files.

1 Source of Truth

The definitive feature sources are:

- base candidate features: `neural/hybrid_model.py`, variable `FEATURE_COLUMNS`;
- JEPA appended feature names: `neural/models/jepa/xinput_v3_production/jepa_feature_names.json`;
- production direction feature order: per-ticker joblib artifacts under `neural/models/jepa/jepa_production_final_180m/base_jepa`;
- option candidate feature definitions: `neural/jepa/train_backtest_option_policy.py`;
- OptionValueJEPA feature inference: `neural/jepa/train_option_value_jepa.py`.

2 Feature Counts

Feature object	Count
Base candidate features in loader	182
Base features present in production <code>base_jepa</code> artifact	180
JEPA appended features	42
Production <code>base_jepa</code> GBT feature columns	222
XInputJEPA state branch	155
XInputJEPA input branch	25
OptionValueJEPA market feature block	222
OptionValueJEPA option feature block	31
OptionValueJEPA dynamic feature block	15

Table 1: Feature counts from the current code and artifacts.

The two base candidate features absent from the final production `base_jepa` artifact are `charm_0dte_vs_wk` and `zomma_0dte_vs_wk`, because they were not present in the final production training table.

3 Key Engineered Market Features

3.1 Net Greek Exposure

These are the core state variables. The important features are:

Feature	Interpretation
<code>net_gamma</code>	Aggregate gamma exposure. It describes how dealer delta changes with spot. The sign and magnitude condition whether price movement is expected to be dampened or amplified.
<code>net_vanna</code>	Aggregate vanna exposure. It connects volatility movement to delta hedging pressure. It becomes especially relevant when IV is moving.
<code>net_charm</code>	Aggregate charm exposure. It captures delta decay through time, particularly important for 0DTE because time decay is intraday and fast.
<code>net_dgex</code>	Delta-gamma exposure proxy. It is used as a directional hedging-pressure descriptor and as a source of DGEX magnet distances.
<code>net_zomma</code>	Gamma sensitivity to volatility. It helps distinguish gamma regimes that are stable from regimes where volatility changes can reshape gamma quickly.
<code>net_delta</code>	Aggregate directional option exposure. It is a direct state descriptor but is most useful in interaction with gamma and vanna.
<code>net_vega</code>	Aggregate volatility exposure. It conditions option payoff and volatility sensitivity.
<code>net_vomma</code>	Convexity of vega. It describes whether volatility exposure itself changes sharply as IV moves.

3.2 Weekly Exposure Features

The weekly fields mirror the 0DTE fields:

```
wk_net_gamma, wk_net_vanna, wk_net_charm, wk_net_dgex, wk_net_zomma, wk_net_delta,  
wk_net_vega, wk_net_vomma.
```

They exist because 0DTE exposure alone can be too local. Weekly exposure can define broader support/resistance or magnetic zones. Ratio fields such as `gamma_0dte_vs_wk`, `vanna_0dte_vs_wk`, `dgex_0dte_vs_wk`, `delta_0dte_vs_wk`, `vega_0dte_vs_wk`, and `vomma_0dte_vs_wk` compare short-expiry pressure to the wider expiry structure.

3.3 Regime and Boolean State Features

Examples:

`gamma_regime, vanna_bullish, charm_bullish, dgex_sticky, zomma_stabilizing,
wk_gamma_regime, wk_vanna_bullish, wk_dgex_sticky, wk_zomma_stabilizing.`

These are not meant to be final trade rules. They are compact regime descriptors. A GBT can use them to learn different behavior in positive/negative gamma, sticky DGEX, or stabilizing zomma environments.

4 Location and Strike Geometry Features

4.1 Distances to Option-Exposure Levels

Important level-distance fields include:

`dist_to_max_gamma, dist_to_min_gamma, dist_to_min_vanna, dist_to_zero_gamma,
dist_to_max_dgex, dist_to_min_dgex, dist_to_max_vega, dist_to_min_vega,
dist_to_max_vomma, dist_to_min_vomma.`

The intuition is spatial. A market near a large max-gamma or DGEX level may behave differently from a market far away from all large option concentrations.

4.2 Initial-Balance Features

The initial-balance family includes:

`price_vs_ib_high, price_vs_ib_low, ib_range_pct, near_ib_high, near_ib_low,
above_ib, below_ib, in_ib_range.`

These features encode whether the current price is inside balance, breaking out, rejecting, or extending. Prior-day variants `dist_ib_high_D1` through `dist_ib_high_D5`, `dist_ib_low_D1` through `dist_ib_low_D5`, and `prev_close_vs_ib_D1` through `prev_close_vs_ib_D5` add multi-day context.

4.3 Fibonacci and Level Features

The model includes extension distances such as:

`dist_fib_127_up, dist_fib_161_up, dist_fib_200_up, dist_fib_127_dn,
dist_fib_161_dn, dist_fib_200_dn.`

It also includes support/resistance style variables:

`nearest_level_dist, level_cluster_density, rejection_bullish, rejection_bearish,
trend_grind_up, trend_flush_down, bouncing_from_support, rejecting_resistance,
wall_at_fib, wall_at_ib, is_touching_fib, is_touching_max_gamma,
is_touching_min_gamma, is_touching_max_dgex.`

5 Volatility, Momentum, and Calendar Features

5.1 Volatility State

Key volatility variables:

```
atm_iv, iv_zscore, iv_percentile, vix_spot, vix_gamma, vix_regime, vix_5d_mean,  
vix_5d_std, atr_5d_norm, rvol_iv_log, rvol_trend, rvol_regime.
```

These features make the model conditional on option richness, realized/expected volatility, and volume/volatility regime.

5.2 Return and Momentum State

Return and momentum features include:

```
ret_1m_vol_adj, ret_5m_vol_adj, ret_15m_vol_adj, momentum_5m_bps, spot_change,  
gamma_momentum, rsi.
```

They allow the model to distinguish a level touch during strong momentum from a level touch during mean reversion.

5.3 Calendar and Time of Day

Time features include:

```
time_sin, time_cos, minutes_to_close_norm, dow_sin, dow_cos, days_to_opex_norm,  
is_opex_week, is_quarterly_opex_week.
```

Time encoding matters because 0DTE exposure, liquidity, and theta decay are highly intra-day.

6 Interaction Features

6.1 Greek Ratios

Ratio features include:

```
gamma_vanna_ratio, dgex_gamma_ratio, charm_vanna_ratio, delta_gamma_ratio,  
vega_gamma_ratio, vomma_vega_ratio.
```

They describe which Greek channel is dominant. For example, large gamma with small vanna is a different state from large vanna with small gamma.

6.2 Confluence Families

Confluence examples:

```
confluence_ib_high_max_gamma, confluence_ib_low_min_gamma,  
confluence_ib_high_max_vega, confluence_ib_low_max_dgex,  
confluence_fib127_bull_max_gamma, confluence_fib161_bull_max_vega,  
confluence_fib127_bear_min_gamma, confluence_fib161_bear_max_vomma.
```

There are also prior-day initial-balance confluences:

```
confluence_dlibh_max_gamma, confluence_dlibl_min_gamma,  
confluence_dlibh_max_dgex, confluence_dlibl_min_dgex, confluence_dlibh_max_vomma.
```

These features encode multi-condition events directly.

7 JEPA Input Features

XInputJEPA input features are designed to behave like exogenous actions or perturbations. The current input candidates are:

Input group	Features
Greek changes	gamma_change, vanna_change, dgex_change, delta_change, vega_change, vomma_change.
Price and momentum changes	spot_change, gamma_momentum, momentum_5m_bps, ret_1m_vol_adj, ret_5m_vol_adj, ret_15m_vol_adj.
Cross-market changes	tlr_ret_1m, tlr_ret_5m, tlr_ret_15m.
Exposure dynamics	gamma_speed, charm_accel_weighted, gamma_phase_delta, pcr_derivative_5m.
Volume and vol perturbations	rvol_trend, rvol_regime, vol_relative, iv_zscore, iv_percentile, vix_5d_std.
Persistence	signal_persistence_5m.

The production artifact contains 25 input features because only columns present in both the parquet and base feature set are used.

8 JEPA Output Features

8.1 Latent State

```
xjepa_z_00, xjepa_z_01, xjepa_z_02, xjepa_z_03, xjepa_z_04, xjepa_z_05, xjepa_z_06,  
xjepa_z_07, xjepa_z_08, xjepa_z_09, xjepa_z_10, xjepa_z_11, xjepa_z_12, xjepa_z_13,  
xjepa_z_14, xjepa_z_15.
```

These are 16 coordinates of the encoded market state.

8.2 Latent Input

xjepa_u_00, xjepa_u_01, xjepa_u_02, xjepa_u_03, xjepa_u_04, xjepa_u_05, xjepa_u_06, xjepa_u_07, xjepa_u_08, xjepa_u_09, xjepa_u_10, xjepa_u_11.

These are 12 coordinates of the recent perturbation/input sequence.

8.3 Derived JEPA Diagnostics

Feature	Meaning
xjepa_latent_velocity	Norm of the change in z from the previous valid context.
xjepa_input_velocity	Norm of the change in u from the previous valid context.
xjepa_pred_dispersion_short	Pairwise dispersion among predicted latent states for horizons 1, 3, and 6.
xjepa_pred_dispersion_long	Pairwise dispersion among predicted latent states for horizons 12, 24, and 36.
xjepa_lagged_pred_30m_err	Error between the 30 minute old latent forecast and the current latent state.
xjepa_lagged_pred_60m_err	Error between the 60 minute old latent forecast and the current latent state.
xjepa_lagged_pred_180m_err	Error between the 180 minute old latent forecast and the current latent state.
xjepa_prob_short	JEPA auxiliary head probability of short regime.
xjepa_prob_hold	JEPA auxiliary head probability of hold/no-trade regime.
xjepa_prob_long	JEPA auxiliary head probability of long regime.
xjepa_trade_confidence	Maximum of short and long probabilities.
xjepa_direction_score	Long probability minus short probability.
xjepa_entropy	Entropy of the three-class auxiliary distribution.
xjepa_context_valid	Indicator that enough history existed to produce a valid JEPA context.

9 Option Candidate Features

The fixed-delta policy does not need a learned option-value feature matrix. However, the research models use the following candidate families:

Family	Features
Ticker and side	ticker_SPX, ticker_QQQ, ticker_SPY, side_LONG, side_SHORT.
Signal state	jepa180_prob_up, jepa180_confidence, jepa180_direction, jepa180_edge.

Family	Features
Time state	<code>entry_minute, pos_in_day, minutes_to_close.</code>
Delta/strike	<code>delta_target, actual_delta, actual_delta_abs, actual_strike, strike_distance_pts, strike_distance_bps.</code>
Premium/spread	<code>entry_premium, raw_entry_premium, entry_spread_pct, premium_to_spot_bps, entry_cost_dollars, contracts.</code>
Option Greeks	<code>actual_iv, actual_theta, actual_gamma, theta_over_premium, gamma_notional.</code>

10 Dynamic Option Exit Features

Learned exit experiments use:

```
hold_minutes, hold_norm, current_pnl_pct, current_return_on_risk, current_premium,
premium_ratio, peak_pnl_pct, drawdown_from_peak, mae_pnl_pct, current_delta,
current_delta_abs, current_iv, current_gamma, spot_return_bps,
signed_spot_return_bps, minutes_remaining.
```

The OptionValueJEPA feature inference excludes `current_return_on_risk` from the dynamic feature block and uses it instead in the exit decision comparison.

11 Feature Interpretation Caveats

1. A feature name can be economically meaningful without being individually predictive.
2. A JEPA latent coordinate can be predictive without being human-interpretable.
3. The GBT may ignore most features. That is acceptable if the retained splits improve OOS performance.
4. Option candidate labels are noisier than direction labels because option payoff includes spread, theta, IV, gamma, and liquidity.
5. Oracle-derived fields must never enter deployable features.

12 Conclusion

The key feature blocks are the net Greek state, strike geometry, volatility regime, intraday location, JEPA latent state/input, and option candidate path state. The evidence supports using JEPA features as a complement to the base GBT features. The next feature-development frontier is not more static market columns; it is better live-safe option-path and liquidity descriptors that can help a learned option selector beat fixed delta.